

Supplemental material for XYZ

1 SUMMA details

At CSSL, the peak water equivalent was XYZ higher in the open site relative to the forest site leading up to the heatwave event which may have resulted from a combination of snow interception or antecedent melt. We found good correspondence between SUMMA and observations in the open and forest sites, respectively. This was found after adjusting the rain/snow threshold temperature to XYZ, the canopy wind reduction coefficient to and a snow surface roughness of 0.002 m for both sites. These values are well within established bounds for those parameters (CITE). Comparisons against the forest site CNR4 radiometer showed good correspondence between the SUMMA modelled below-canopy shortwave radiation, lending confidence to the prescribed LAI value and UEB 2-stream shortwave parameterization implemented in SUMMA.

2 Additional details about SUMMA

The bulk expression for the H_s implemented in SUMMA is given by:

$$H_s = c_p \rho C_s u (T_{air} - T_{snow}), \quad (1)$$

where c_p is the specific heat capacity of dry air (constant), u is the horizontal wind speed, ρ is the dry air density, and C_s is the bulk transport coefficient. In this work, we use the SUMMA formulation for C_s , which models the stability correction as a function of the bulk Richardson number.

We use the “Loisinv” stability correction function implemented in the SUMMA model and assume that $z_h = z_0$

$$C_h = \frac{k^2}{\ln(z/z_0)^2} \cdot f(\text{Ri}) \quad (2)$$

Where Ri is the bulk Richardson number, defined as

$$\text{Ri} = \frac{9.81(T_{air} - T_{snow})}{0.5(T_{air} + T_{snow})u^2}. \quad (3)$$

and $f(\text{Ri})$ is a function with range (0,1] that declines as a function of increasing Ri.

3 Validation

Stuff here.

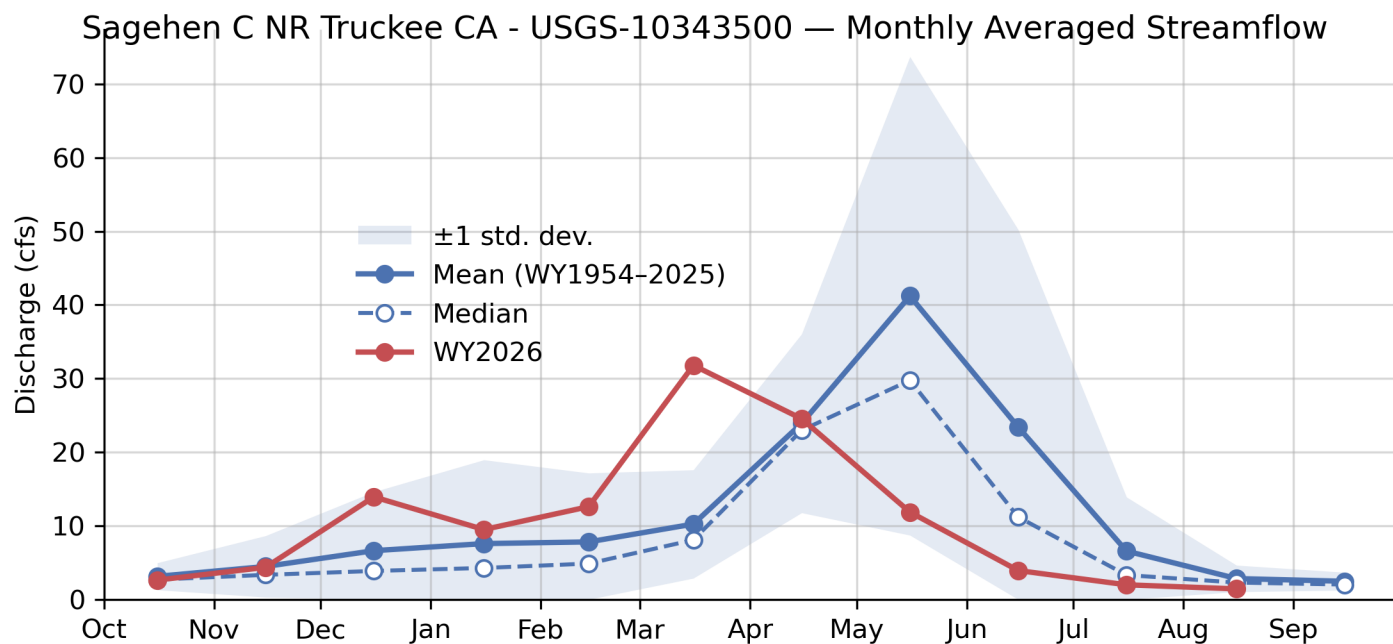


Figure S1: The monthly averaged streamflow anomalies in cubic-feet-per second (cfs) between 1954 to present for the USGS Sagehen Creek gage site. The historical mean, historical median, historical 1-standard deviation (shaded region), and WY2026 streamflow are depicted.

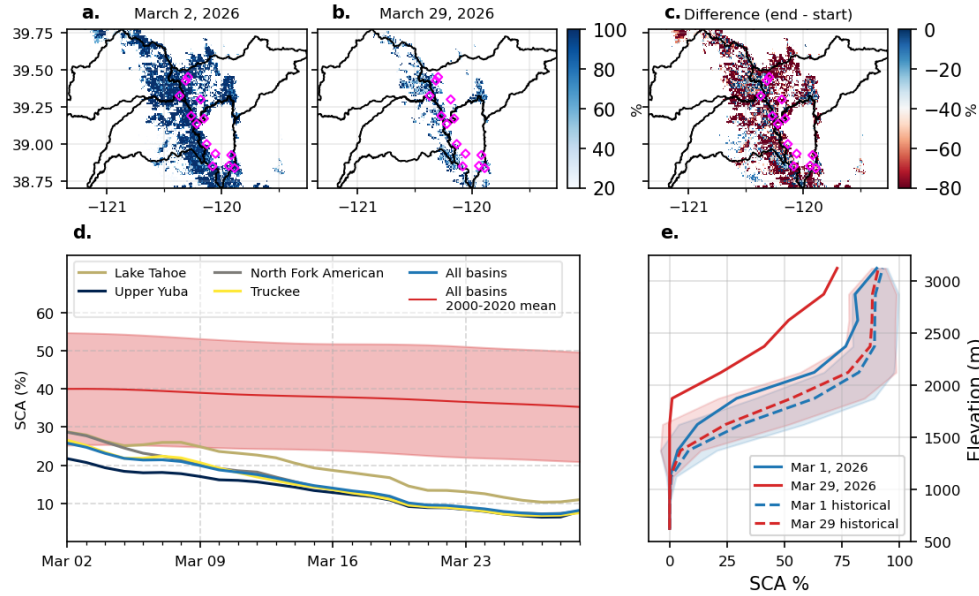


Figure S2: SPIRES SCA across the study region during the March heatwave. Watershed boundaries shown in black and SNOTEL monitoring sites indicated by magenta diamonds. **(a, b, c)** SCA (%) on March 2 and March 29, 2026, respectively, and the difference between the two dates. **(d)** Daily mean SCA (%) for each individual watershed and aggregated across all four basins. The 2000–2020 historical mean and ± 1 standard deviation are depicted. **(e)** Mean SCA as a function of elevation band (250 m bins) for March 2 (blue) and March 29 (red), shown alongside historical values for the same dates (dashed lines) with ± 1 standard deviation shading.

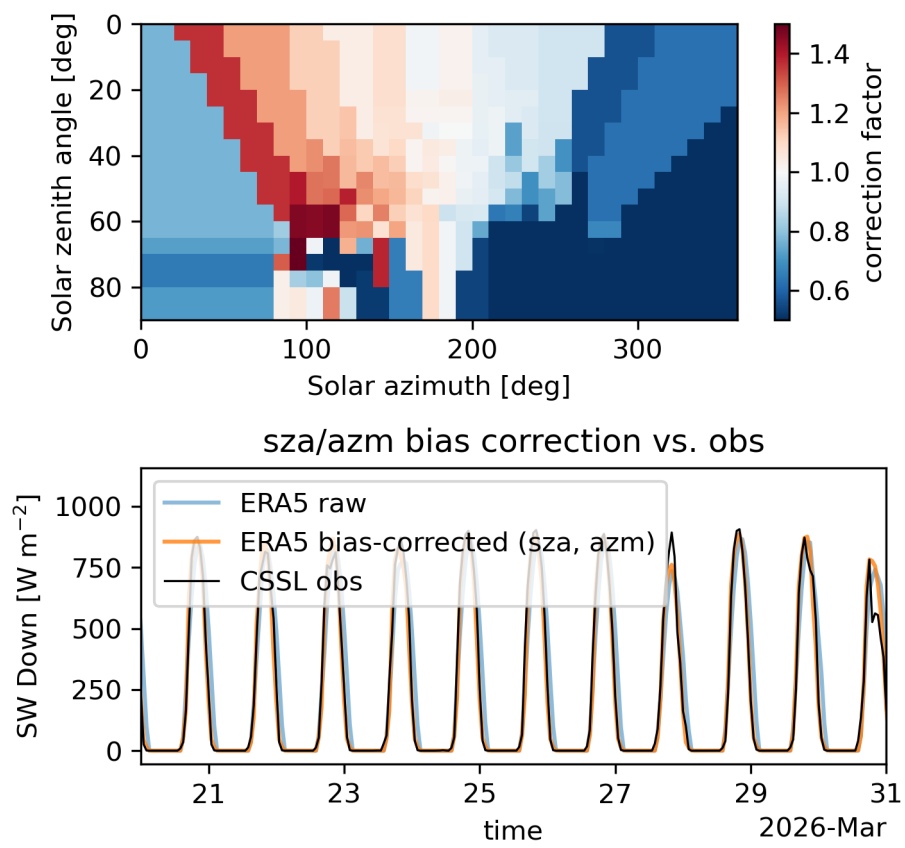


Figure S3:

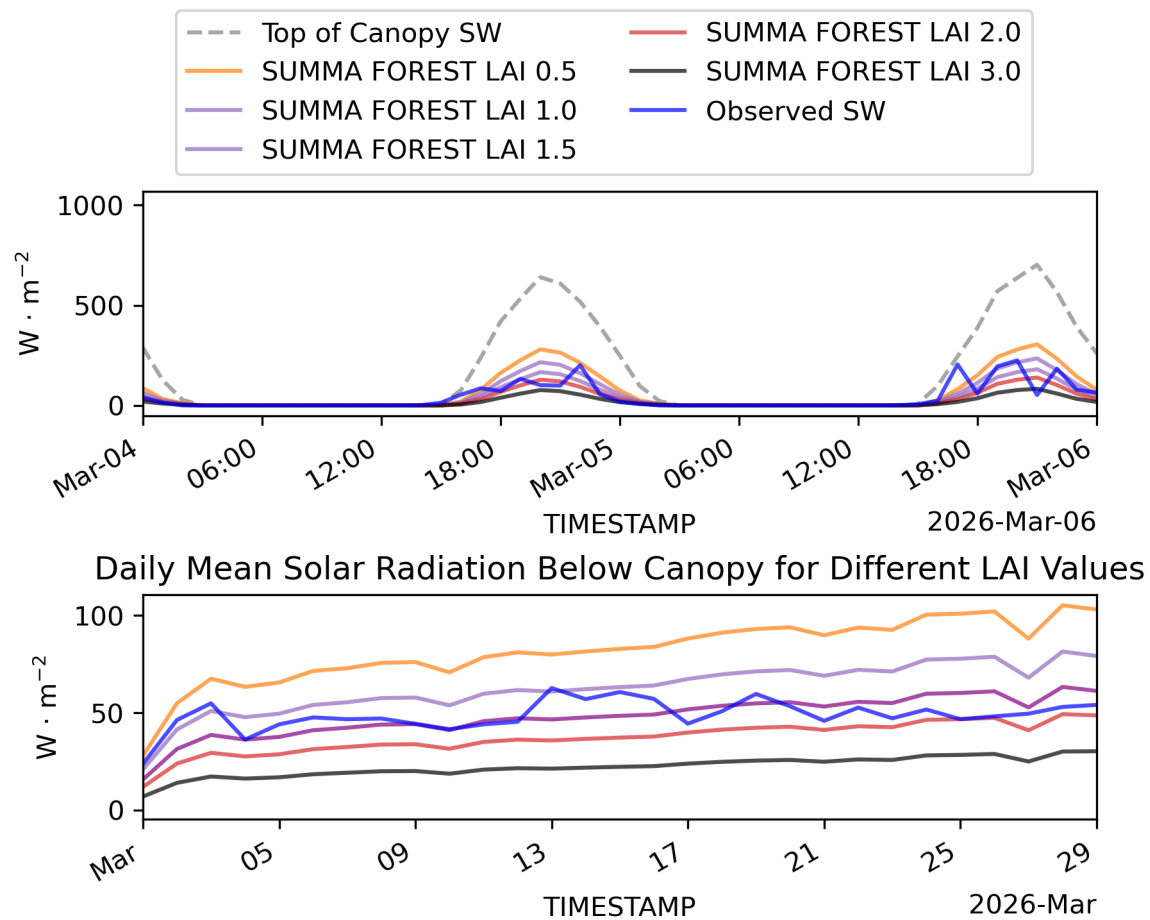


Figure S4: A comparison of sub-canopy downwelling radiation for different LAI values using the UEB-2stream shortwave radiation parameterization. The top plot depicts a hourly $SW_{\downarrow}$ two day period in early March. The bottom plot depicts hourly $SW_{\downarrow}$ during March. Observed radiation by the Kipp and Zonen radiometer is depicted

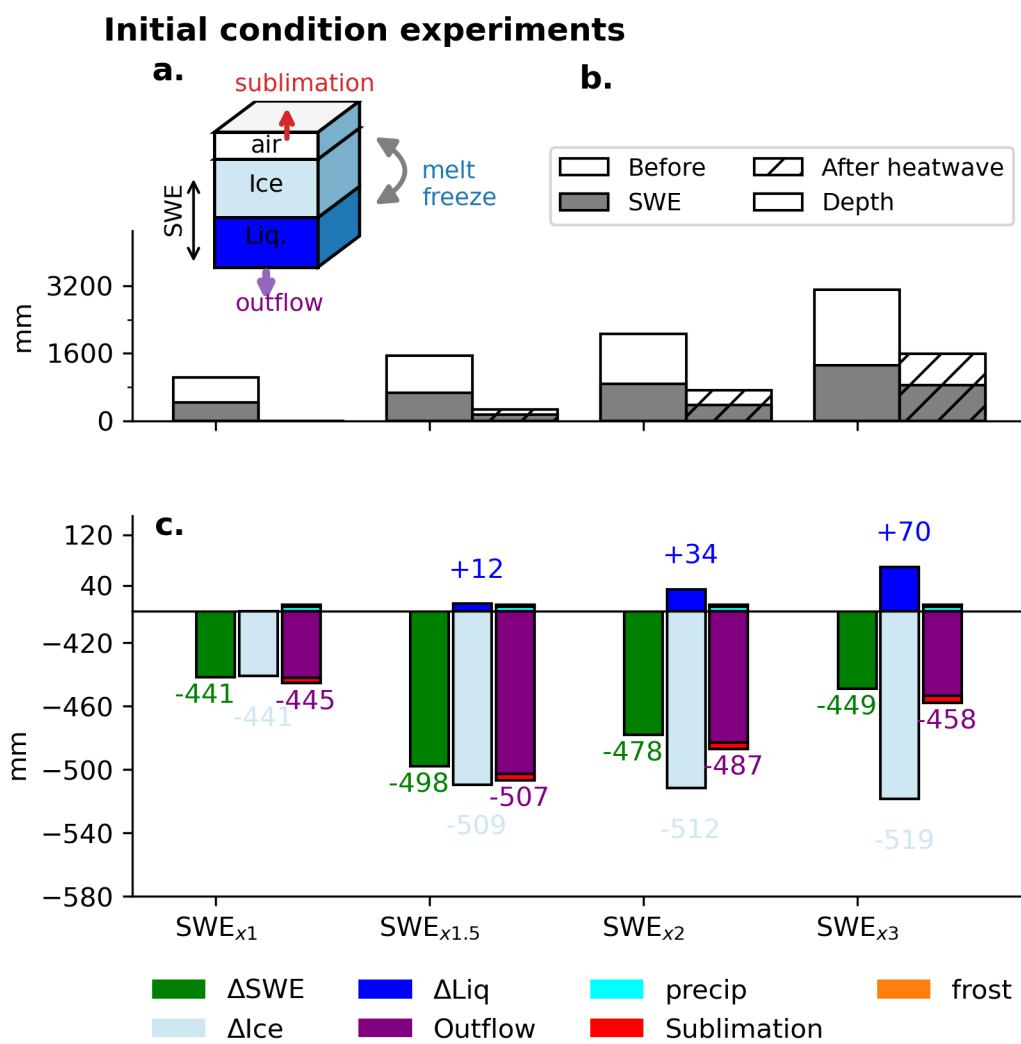


Figure S5: Initial condition tests